

**Faculty Development
Program - Day**

05

Organised by Groware Education
Solutions, Principal Affiliate of
Kerala State Rutronix



കേരള സ്റ്റേറ്റ് റൂട്രോണിക്സ്



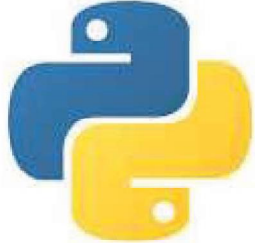


Python List as Stack

- To add an item to the top of the stack, use **append()**. To retrieve an item from the top of the stack, use **pop()** without an explicit index.

Example Program

```
#Demo of List as Stack  
stack=[10,20,30,40,50]  
stack.append(60)  
print("Stack after appending:",stack)  
stack.pop()  
print("Stack after popping:",stack)
```



Python List as QUEUE

To implement a queue, Python provides a module called **collections** in which a method called **deque** is designed to have fast appends and pops from both ends.

Example Program

#Demo of List as Queue

```
from collections import deque
```

```
queue = deque(["apple", "orange", "pear"])
```

```
queue.append("cherry")           # cherry arrives
```

```
queue.append("grapes")          # grapes arrives
```

```
queue.popleft()                 # The first to arrive now leaves
```

```
queue.popleft()                 # The second to arrive now leaves
```

```
print(queue)                   # Remaining queue in order of arrival
```



Python Tuple

- A tuple is another sequence data type that is similar to the list.
- A tuple consists of a number of values separated by commas.
- The main differences between lists and tuples are lists are enclosed in square brackets ([]) and their elements and size can be changed, while tuples are enclosed in parentheses (()) and cannot be updated.
- Tuples can be considered as read-only lists.

Example

(1, 'Ram', 120, 12.9)



Python Tuple

Example Program

```
first_tuple = ('abcd', 147, 2.43, 'Tom', 74.9)
small_tuple = (111, 'Tom')
print(first_tuple)
print(first_tuple[0])
Print(first_tuple[-1])
print(first_tuple[1:3])
print(first_tuple[2:])
print(small_tuple * 2 )
print(first_tuple + small_tuple)
```



Python Tuple – Built-in Functions

- **len(tuple)** – Gives the total length of the tuple.
- **max(tuple)** – Returns item from the tuple with maximum value.
- **min(tuple)** – Returns item from the tuple with minimum value.
- **tuple(seq)** – Returns a list into a tuple.